

1 Impulse Representation

(Sifting Property) The sifting property is the ability of the Dirac delta function to “pick out” the value of a function at a specific point during integration.

1.1 DT

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

1.2 CT

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau) d\tau = x(t) * \delta(t)$$

2 Impulse Response

Let S be an *LTI system* and $x[n] \xrightarrow{S} y[n]$.

Suppose $\delta[n] \xrightarrow{S} h[n]$ and $\delta[n-k] \xrightarrow{S} h_k[n]$. Since S is time-invariant, we have $h_k[n] = h[n-k]$. Together with the linearity, we have

$$\begin{aligned} x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] &\xrightarrow{S} y[n] = \sum_{k=-\infty}^{\infty} x[k]h_k[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= x[n] * h[n] \end{aligned}$$

Remark. In the impulse representation $x[n] * \delta[n]$, $x[n]$ becomes a constant sequence of scalars, and $\delta[n]$ becomes the one that actually carries the signal.

The same for continuous-time:

$$\begin{aligned} x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau) d\tau &\xrightarrow{S} y(t) = \int_{-\infty}^{\infty} x(\tau)h_{\tau}(t) d\tau \\ &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau \\ &= x(t) * h(t) \end{aligned}$$